Lessons 7-8: Student Research Project

Unit

Working Like a Physicist: Collecting, Processing, and Presenting Data

Lesson focus

Students work in pairs or groups of 3 to design, carry out, present, and evaluate their own investigation. They choose a research question in the form "How does X affect Y?" and apply the data handling skills developed across the unit.

Project title

Student research project: How does X affect Y?

Project aim

Students independently apply the full investigation process:

1. Choose a suitable research question.
2. Identify independent, dependent, and control variables.
3. Decide whether the independent variable is continuous or categoric.
4. Design a fair test.
5. Collect repeated measurements.
6. Calculate mean values.
7. Choose and draw the correct graph.
8. Draw a best-fit line where appropriate.
9. Identify possible anomalous results.
10. Write a conclusion and evaluation based on evidence.

Assumptions

Students have completed Lessons 1-6 and have practiced judging whether data can be trusted, variables, data types, graph choice, repeated measurements, means, bar charts, line graphs, best-fit lines, anomalous results, conclusions, and evaluations.

Students can work in pairs or groups of 3. Groups of 3 are useful where equipment is limited or where one student can act as a timer or recorder.

Learning objectives

By the end of Lessons 7-8, students should be able to:

1. Write a testable research question in the form "How does X affect Y?"
2. Plan a safe and fair investigation.
3. Identify independent, dependent, and control variables.
4. Collect repeated measurements in a clear results table.
5. Calculate mean values accurately.
6. Choose a bar chart or line graph using the type of independent variable.
7. Draw a high-quality graph with labels, units, sensible scale, and accurate plotting.
8. Use a best-fit line where appropriate.
9. Comment on anomalous results, errors, and improvements.

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10 Present a conclusion supported by data.

Key vocabulary

Keyword	Student-friendly definition
Research question	The question the investigation is trying to answer
Independent variable	The thing you change
Dependent variable	The thing you measure
Control variable	Something you keep the same
Continuous data	Number data that can have many possible values
Categoric data	Data sorted into groups or types
Trial	One repeat of a measurement
Mean	The average value
Bar chart	A graph used when the independent variable is categoric
Line graph	A graph used when the independent variable is continuous
Best-fit line	A line that shows the overall pattern of the results
Anomalous result	A result that does not fit the pattern
Random error	Unpredictable variation between repeats
Systematic error	A repeated error that shifts all results in the same way

Equipment and resources

Possible equipment, depending on available practical choices:

- Ramps
- Trolleys or toy cars
- Meter rulers or tape measures
- Stopwatches
- Balls or marbles
- Paper helicopter templates
- Paperclips
- Elastic bands
- Springs
- Masses
- Different surfaces or materials
- Graph paper
- Exercise books or project sheets
- Calculators
- Safety goggles, if needed for the chosen practical

Safety notes

- Students must have their research question and method checked by the teacher before collecting data.
- Students should not climb on chairs or benches.

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- Moving objects must not be launched toward people.
- Heavy objects should be handled carefully.
- Scissors should be used carefully.
- Groups should keep bags, stools, and loose equipment away from the testing area.
- Any investigation involving heat, chemicals, electricity, sharp objects, or high drops should only be used if specifically approved by the teacher.

Suitable research question format

All groups must write their research question in this format:

> *How does X affect Y?*

For example:

Research question	Independent variable	Dependent variable	Likely graph
How does ramp height affect distance travelled by a trolley?	Ramp height	Distance travelled	Line graph
How does wing length affect fall time of a paper helicopter?	Wing length	Fall time	Line graph
How does surface type affect distance travelled by a toy car?	Surface type	Distance travelled	Bar chart
How does material type affect bounce height of a ball?	Material type	Bounce height	Bar chart
How does number of paperclips affect fall time of a paper helicopter?	Number of paperclips	Fall time	Line graph or bar chart, depending on how it is treated

Unsuitable research questions

Students may need help rejecting questions that are too vague, unsafe, or not measurable.

Unsuitable question	Problem
What is the best paper helicopter?	Too vague
How does a car move?	Not a fair-test investigation
How does height affect it?	Dependent variable is unclear
How does surface affect speed and distance and time?	Too many dependent variables
How does throwing force affect distance?	Difficult to control fairly

Project rules

Each group must:

1. Work in a pair or group of 3.
2. Choose one research question in the form "How does X affect Y?"
3. Change only one independent variable.
4. Measure one dependent variable.
5. Identify at least 3 control variables.
6. Test at least 4 or 5 values or categories of the independent variable.

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7. Collect at least 3 trials for each value or category.
8. Calculate a mean for each value or category.
9. Choose the correct graph type.
10. Write a conclusion and evaluation.

Lesson 7: Planning and Data Collection

Starter: What makes a good research question?

Show students these examples:

1. How does ramp height affect distance travelled?
2. How does material type affect bounce height?
3. How does wing length affect fall time?
4. How does the number of elastic bands affect launch distance?

Ask students:

1. What is the independent variable?
2. What is the dependent variable?
3. Is the independent variable continuous or categoric?
4. What graph would be suitable?
5. What would need to be kept the same?

Activity 1: Choose a research question

Students work in pairs or groups of 3 to choose a research question.

They should complete the planning grid before starting practical work.

Planning question	Student answer
What is your research question?	
What is the independent variable?	
What is the dependent variable?	
Is the independent variable continuous or categoric?	
What graph will you use?	
Why is that graph suitable?	
What are at least 3 control variables?	
What equipment will you need?	
What safety risks are there?	
How will you reduce those risks?	

Activity 2: Method design

Students write a short method before collecting data.

Their method should include:

1. The values or categories of the independent variable.
2. How the dependent variable will be measured.

3. How many trials will be completed.
4. How control variables will be kept the same.
5. How the investigation will be carried out safely.

Teacher checkpoint

Before students collect data, the teacher should check:

Check	Yes/No
Research question is clear and testable	
Independent variable is clear	
Dependent variable is clear and measurable	
Control variables are realistic	
Method is safe	
Equipment is available	
Repeats are included	
Results table has units	
Graph choice is sensible	

Activity 3: Results table

Students design their own results table. A support version is provided below.

Independent variable / unit	Trial 1 / unit	Trial 2 / unit	Trial 3 / unit	Mean / unit

For categoric independent variables, students should replace "Independent variable / unit" with the category name, for example "Material type" or "Surface type".

Activity 4: Data collection

Students carry out their investigation and collect repeated measurements.

During data collection, students should:

1. Record results immediately.
2. Include units in the table headings.
3. Repeat each measurement at least 3 times.
4. Check any unusual result before moving on.
5. Keep control variables the same.
6. Work safely and responsibly.

Lesson 7 plenary

Students complete:

1. Our research question is _____.

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2. Our independent variable is _____.
3. Our dependent variable is _____.
4. One control variable we kept the same was _____.
5. One result we may need to check is _____.

Lesson 8: Graphing, Analysis, and Presentation

Starter: Which graph should we use?

Students look back at their independent variable and answer:

1. Is it continuous or categoric?
2. Should the graph be a line graph or bar chart?
3. What should go on the x-axis?
4. What should go on the y-axis?
5. What units are needed?

Key rule:

> *Use a line graph when the independent variable is continuous.*
> *Use a bar chart when the independent variable is categoric.*

Activity 1: Calculate means

Students calculate the mean for each value or category of the independent variable.

They should show at least one example calculation:

> *Mean = total of the trials / number of trials*

Activity 2: Draw the graph

Students draw the correct graph for their data.

Graph requirements:

1. Clear title.
2. Independent variable on the x-axis.
3. Dependent variable on the y-axis.
4. Units included where needed.
5. Sensible scale.
6. Accurate plotting.
7. Bars with gaps if drawing a bar chart.
8. Small crosses if drawing a line graph.
9. Best-fit line if drawing a line graph.
10. Possible anomalous results circled or commented on.

Activity 3: Analyse the results

Students answer:

1. What pattern does the graph show?

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2. Does the dependent variable increase, decrease, or stay about the same?
3. Is the relationship strong or weak?
4. Are there any possible anomalous results?
5. What could have caused random error?
6. What could have caused systematic error?

Activity 4: Final project page

Each group produces one final project page containing:

1. Research question.
2. Variables.
3. Method summary.
4. Results table.
5. Mean values.
6. Correct graph.
7. Comment on anomalous results or errors.
8. Conclusion.
9. Evaluation.

Conclusion structure

Students write a conclusion using this structure:

> *As _____ increased/changed, _____ increased/decreased/stayed about the same.*
> *I know this because _____.*
> *The graph shows _____.*
> *One result that may be an anomalous result is _____ because _____.*

For a categoric investigation, students can use:

> *The category with the highest/lowest _____ was _____.*
> *I know this because _____.*
> *This suggests that _____.*

Evaluation structure

Students write an evaluation using this structure:

> *One strength of our method was _____.*
> *One weakness of our method was _____.*
> *One possible random error was _____.*
> *One possible systematic error was _____.*
> *If we repeated the investigation, we would improve it by _____.*

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Peer review checklist

Students swap work with another group and check:

Feature	Yes/No
Research question uses "How does X affect Y?"	
Independent variable identified	
Dependent variable identified	
At least 3 control variables identified	
Results table includes units	
At least 3 trials included	
Mean values calculated	
Correct graph type chosen	
Axes labeled correctly	
Units included on graph	
Sensible scale used	
Points or bars plotted accurately	
Best-fit line drawn where appropriate	
Anomalous results considered	
Conclusion uses evidence	
Evaluation includes an improvement	

Success criteria

Skill	Success criteria
Research question	Clear, testable, and written as "How does X affect Y?"
Variables	Independent, dependent, and control variables identified correctly
Method	Safe, repeatable, and controls key variables
Results table	Includes headings, units, trials, and mean values
Repeats	At least 3 trials for each value or category
Graph choice	Correct graph chosen using the independent variable
Graph quality	Title, axes, units, sensible scale, and accurate plotting included
Best-fit line	Used for line graphs where appropriate
Anomalous results	Possible anomalous results identified or absence of anomalous results stated
Conclusion	Describes the pattern and uses evidence from results
Evaluation	Identifies errors, weaknesses, and a realistic improvement

Optional extension

More confident students can:

1. Compare their results with another group.
2. Explain whether their data is repeatable.

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3. Suggest a wider range of values for the independent variable.
4. Explain whether the pattern is linear or non-linear.
5. Add error bars if appropriate.

Plenary

Students complete:

1. The most important decision in our investigation was _____.
2. The strongest evidence from our results was _____.
3. The biggest source of error was probably _____.
4. If we did this again, we would improve _____.

Possible prep

Students write an individual reflection:

1. What did your group investigate?
2. What did your graph show?
3. What did you personally contribute?
4. What was the most difficult part of the project?
5. What would you do better next time?

Teacher notes

This two-lesson project is intended as a light-touch assessment of the whole unit. The main priority is not perfect data, but whether students can think and communicate like physicists: choose variables carefully, collect repeated measurements, present data clearly, and make a conclusion that follows from evidence.

Some groups may choose categoric independent variables and produce bar charts, while others may choose continuous independent variables and produce line graphs. This is useful because it allows the teacher to check whether students understand that graph choice depends mainly on the independent variable.

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